ÇANKAYA UNIVERSITY

COMPUTER ENGINEERING DEPARTMENT

CENG 407

Final Report

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0. Abstract

The use of virtual/mixed reality and patient-specific three-dimensional models in medical education and neurosurgical applications has been rapidly increasing in recent years. In parallel, automated and precise anatomical modeling has become possible thanks to deep learning-based brain MR segmentation. This study reviews the current literature on (3) VR/MR-based anatomy and medical education, (3.1) deep learning-based brain tumor segmentation, and (3.2) the use of patient-specific 3D models in VR/MR environments for neurosurgical planning and patient information; it discusses the strengths and weaknesses of existing studies and reveals the main research gaps in the field.

Keywords: Brain MR, deep learning, segmentation, virtual reality, mixed reality, neurosurgery, medical education, patient-specific model.

1. Introduction

Magnetic Resonance Imaging (MRI) is one of the fundamental diagnostic methods that allows for the non-invasive examination of brain structures with high soft tissue resolution [1]. In clinical practice, MRI data are mostly evaluated as 2D slices, and physicians attempt to understand the 3D anatomical structure and the location of pathological lesions by mentally synthesizing these slices. Especially when dealing with the skull base, deep-seated tumors, and complex vascular pathologies, this mental 3D reconstruction process constitutes a significant cognitive load and becomes a highly experience-dependent process [2], [3].

Similarly, medical education, particularly neurosurgery training, has for many years been conducted through cadaver dissection, 2D atlases, and flat screen images. While these methods provide fundamental anatomical knowledge, they remain limited in understanding complex 3D relationships and patient-specific variations [4]. Although Virtual Reality (VR) and Mixed Reality (MR)-based training applications have been developed in recent years, carrying the potential to increase students' spatial perception and learning motivation [5], a significant portion of these applications are based on standard, idealized anatomical models. Consequently, it is often not fully possible to reflect the anatomical and pathological features specific to a clinical case in the training and planning process.

In this context, automatically generating 3D, patient-specific brain models from 2D MRI slices and interactively visualizing these models in a VR/MR environment presents a significant opportunity for both medical education and neurosurgical operation planning. Deep learning-based segmentation methods allow for the automatic and highly accurate separation of brain tumors and other intracranial structures on MRI [6], [7]; these segmentation outputs can then be converted into 3D surface models through appropriate processing steps. However, the literature reveals that the number of portable systems, focused on education and planning, that combine this AI-based segmentation process within an end-to-end workflow: 2D MRI -> 3D patient-specific model -> VR/MR-based interactive visualization is quite limited [8].

The objective of this study is to create 3D patient-specific models through deep learning-supported automatic segmentation from brain MRI data and to present these models interactively in a virtual/mixed reality environment. The proposed approach is expected to facilitate the process of spatial understanding by reducing the need for mental 3D reconstruction of 2D MRI slices, offer a more realistic and rich learning experience in medical/anatomy education, and provide additional visual support to the clinician during neurosurgical operation planning.

A. Purpose of this Study

The main objective of this study is to develop an integrated system that can automatically create 3D patient-specific models from brain MRI data and present these models interactively within a Virtual/Mixed Reality (VR/MR) environment.

This project addresses a critical gap in current clinical and educational workflows by establishing a single, seamless pipeline from 2D imaging to immersive 3D visualization.

1. **Automated Segmentation:**
   Contribution: Utilizing a deep learning-based method to achieve the automatic segmentation of brain structures and pathological areas from standard 2D MRI slices. This eliminates manual segmentation effort.

2. **3D Model Production:**
   Contribution: Processing the automated segmentation outputs to generate a high-resolution, patient-specific 3D brain model unique to the real patient. This step converts the raw data output into a clinically usable format.

3. **VR/MR Based Interactive visualization:**
   Contribution: Enabling the interactive examination of the 3D model through a portable VR/MR device (like Meta Quest). Interactive tools include rotation, zooming, layer peeling (dissection), and virtual cross-sectioning.

4. **A new approach for Medical Education:**
   New Approach: Provides a novel method for teaching by allowing students to examine complex anatomical relationships in a three-dimensional context using real patient cases.

   Benefit: Overcomes the spatial comprehension limitations inherent in learning brain anatomy from traditional 2D atlases.

5. **Neurosurgical Planning and Patient Information:**
   Planning: Assists the surgeon in achieving a better understanding of the patient-specific pathology and surrounding structures, facilitating precise surgical approach planning.

Patient Education: Allows the surgeon to demonstrate the procedure to the patient in a more comprehensible and visual manner, improving consent and trust.

B. Expected System Outcomes

The successful development of this system is anticipated to:

- Increase Spatial Comprehension in both educational and clinical settings.
- Support Decision-Making Processes for complex surgeries.
- Reduce the Cognitive Load typically associated with mentally reconstructing 3D anatomy from 2D images.

2. Definitions

Term(s)	**Context**
MRI	Magnetic Resonance Imaging (MRI); a radiological technique used to image soft tissues [1].
Segmentation	The process of classifying pixels in digital images as an anatomical structure or pathology (tumor) [4].
VR	Simulation technology where the user is completely isolated from the external world and immersed in a digital environment [11].
MR(Mixed Reality)	Technology in which virtual objects are overlaid onto the real world image and can be interacted with [19].
Dice Score(DSC)	A similarity coefficient that measures the success of automatic segmentation and takes a value between 0 and 1 [5].
U-Net	A deep learning model developed for biomedical image segmentation, which has a 'U'-shaped architecture [15].
Patient-Specific Model	A personalized 3D structure generated from the patient's own MR/CT data, instead of a standard anatomical atlas [7].

3. Reviewed Applications and Models

Recent systematic reviews demonstrate that VR and AR technologies are increasingly being adopted in anatomy education [1], [2]. A comprehensive meta-analysis by García-Robles et al., examining studies between the years 2000–2024, reports that immersive VR and AR-based anatomy applications significantly increase students' knowledge level and spatial perception [1].

The systematic review by Minouei et al. emphasizes that VR increases medical students' anatomy performance, but stresses that the highest effectiveness emerges when VR is used as a complementary addition to traditional methods [2]. Niu et al., in a study where 3D anatomy models were integrated into a VR-supported blended learning framework, report a significant increase in both examination scores and student satisfaction in the groups using VR [9].

Comprehensive meta-analyses examining the effectiveness of VR/AR across health education in general also report that VR-based training offers positive effects, particularly on skill level, motivation, and self-efficacy [10], [11]. However, a significant portion of these studies used ready-made commercial anatomy applications or general anatomical models, and patient-specific models were rarely addressed.

A. Limitations

The common limitations of the studies in this area are summarized as follows:

- Most studies have small sample sizes and are single-center [1] – [9].
- Long-term knowledge retention and transfer to real clinical performance have been rarely measured [10], [11].
- The VR content used mostly consists of generalized, non-patient-specific anatomy models; the use of personalized brain models based on MR/CT is highly limited.

3.1. Deep Learning-Based Brain MR Segmentation

A. Fully Automated Deep Learning Methods

Brain tumor segmentation is one of the most challenging problems in medical image analysis. Despite this difficulty, deep learning methods have achieved significant success, particularly with large-scale datasets like BraTS (Brain Tumor Segmentation Challenge) [4], [5], [12].

In a review by Liu et al., it was shown that Convolutional Neural Network (CNN)-based models, especially U-Net and its variants, demonstrate performance close to or exceeding manual segmentation, capable of automatically isolating the tumor core, edema, and enhancing regions [4]. A more recent prospective review by Abidin et al. provides a detailed classification of CNN, Transformer, and hybrid models that utilize multi-modal MR (T1, T2, FLAIR, T1ce) and reports that Dice scores have reached the 85–90% range in recent years [5].

Dorfner et al. further show that deep learning is used not only for segmentation but also for tumor classification, distinguishing between recurrence and radionecrosis, and predicting molecular biomarkers (such as IDH, MGMT, etc.) [13]. These studies describe a broader clinical decision support chain where segmentation is used as an initial step.

From Turkey, Çay et al. compared two different deep learning architectures (U-Net and SegNet) demonstrating that U-Net offers higher accuracy and generalizability [9]. Similarly, Abdusalomov et al. reported that an MR-based deep learning model can detect tumor presence and localization with high sensitivity and specificity [14].

B. Semi-Supervised and Data Efficiency Approaches

The most significant problem with fully supervised deep learning methods is their dependence on large labeled datasets that require pixel-level annotation. A semi-supervised review by Jin et al. shows that competitive segmentation performance can be achieved with limited labeled data using techniques such as pseudo-labeling, consistency regularization, GAN-based approaches, and contrastive learning [12]. This presents a significant advantage for real-world clinical centers aiming to produce patient-specific models.

C. Limitations

- Most models operate in off-line, research environments; real-time or VR/MR integration generally remains at a hypothetical level [5] – [13].
- Issues concerning generalizability, robustness, explainability, and regulation (CE/FDA, etc.) for clinical use are still significant barriers.
- The majority of studies focus only on the tumor region; integrated segmentation of vascular structures, functional areas (e.g., motor cortex), and white matter tracts is limited.

3.2. Patient-Specific 3D Models and VR/MR-Based Neurosurgical Applications

A. 3D Printing and Digital 3D Models

Panesar et al. demonstrated that patient-specific 3D printed models offer significant added value for preoperative planning and education, especially for skull base tumors and complex vascular pathologies [7]. The models were used by surgeons to practice surgical approaches, visualize critical structures, and for patient/family education before the operation.

An narrative review by Isikay et al. on patient-specific 3D visualization and "reality technologies" (VR/AR/MR) in skull base neurosurgery emphasizes that these technologies increase spatial awareness in surgical training, planning, and navigation. Furthermore, they contribute to the development of surgical approaches that are respectful of critical neurovascular structures [16].

B. Neurosurgical Planning and Patient Information using VR and MR

Studies on patient-specific VR platforms have shown that both surgical strategy and intraoperative navigation can be improved using 360° VR environments generated from preoperative MR/CT data [17], [20]. These platforms enable the virtual testing of different access routes, the avoidance of risky vascular structures, and even the interactive display of the patient's own anatomy to the patient.

Crispi et al. reported a significant increase in patients' understanding of their disease and their confidence in the treatment decision, in a study where MR-based 3D brain visuals were displayed using mixed reality headsets in addition to routine neurosurgical clinic consultations [19].

Colombo et al. and various other groups have reported that mixed reality provides advantages in terms of ergonomics and intuitive understanding compared to classic 2D imaging and classic navigation in cranial neurosurgical planning for tasks such as surgical corridor selection, determining the craniotomy site, and staying clear of vascular structures [18], [20].

Furthermore, studies such as planning patient-specific reconstruction plates using mixed reality systems [19] and AR navigation for arteriovenous malformation surgery [20] also indicate the potential of these technologies to reduce surgical time and error.

3.3. Evaluation Criteria Used In Existing Studies

The studies reviewed in the literature utilize different evaluation metrics according to three main research axes: (A) deep learning-based MRI segmentation, (B) VR/MR-based medical education applications, and (C) the clinical accuracy of patient-specific 3D models and VR-supported surgical planning. This section summarizes the criteria by which the success of the studies in the field is evaluated and highlights the methodological differences between the research areas.

A. Metrics Used in Deep Learning-Based MRI Segmentation Studies

In segmentation research aiming for the automatic extraction of tumors or anatomical structures from MR images, quantitative metrics are widely used to measure accuracy.

- **Dice Similarity Coefficient (DSC):** It is the most commonly used metric to measure segmentation accuracy. It takes a value between 0 and 1, where 1 indicates ideal agreement.
- **Intersection over Union (IoU / Jaccard Index):** It measures the ratio of intersection to union between the predicted region and the ground truth annotation.
- **Sensitivity / Recall:** Indicates how well the model can capture positive classes such as a tumor.
- **Specificity:** Evaluates whether normal tissue is correctly preserved without false positive classification.

- **Hausdorff Distance (HD95):** Measures how close the segmentation boundaries are to the true boundary; this is especially important in preparation for surgery.
- **Voxel-wise Accuracy:** Gives the rate of correct classification for every single pixel; it summarizes the global accuracy.

These metrics establish a common language for comparing deep learning models; furthermore, they allow for the detailed analysis of performance across sub-structures such as the brain tumor, edema, or the contrast-enhancing region.

B. Metrics Used in VR/MR-Based Medical Education Studies

VR/MR-based educational research is mostly evaluated through learning outcomes, user experience, and performance improvement.

- **Student achievement scores (pre-test / post-test):** This is the fundamental criterion used to measure the effect of VR training on knowledge gain.
- **Spatial awareness scales**: Measures the student's level of understanding of 3D relationships.
- **System Usability Scale (SUS):** Evaluates the user experience of the VR application in terms of accessibility, usability, and ergonomics.
- **Self-efficacy and satisfaction surveys:** Measures the student's self-confidence, motivation, and attitude toward the application.
- **Task completion time:** Indicates how efficiently the participant can complete a specific clinical task within the VR environment.

These criteria are used to objectively compare how VR/MR-supported education affects learning performance compared to traditional methods.

C. Metrics Used in Studies on Patient-Specific 3D Models and VR-Supported Neurosurgical Planning

Patient-specific models used in surgical planning are evaluated not only as a visual tool but also in terms of measurable clinical outcomes.

- **Anatomical Accuracy (measurement error – mm):** Evaluates the difference between the produced 3D model and the true anatomical structure; this is of critical importance in skull base surgery.
- **Reduction in surgical time:** The contribution of VR-supported planning to the duration of the operation is measured.
- **Planning Accuracy:** The accuracy of the approach chosen by the surgeon, the craniotomy site, and the navigation path is evaluated.
- **Error rate / Deviation from critical structures:** Shows whether model-supported planning reduces risks.

- **Surgeon satisfaction and ease of use:** This is one of the most critical subjective metrics that determines clinical applicability.

These criteria are used to demonstrate the aspects in which VR/MR-based surgical planning and patient communication approaches provide an advantage compared to traditional 2D image review methods.

4. General Assessment and Synthesis of the Literature

When the reviewed literature is evaluated from a holistic perspective, it is clear that VR/MR-based medical education leads to a significant increase in student learning and spatial awareness; deep learning models achieve high accuracy levels in brain tumor segmentation; and patient-specific 3D models offer valuable contributions to both surgical planning and patient information processes. However, a noticeable point is that these three research areas are progressing largely independently of one another. Most VR applications use standard and idealized anatomical models, while deep learning-based segmentation studies primarily focus only on technical accuracy metrics; and integrated system studies focusing on the interactive use of 3D patient-specific models in VR/MR environments remain highly limited.

Therefore, a critical gap still exists in the literature:

"2D MRI → Automated segmentation → 3D patient-specific model → Interactive VR/MR inspection" The lack of a platform where the steps are combined end-to-end, and which is usable for both education and surgical planning.

This study aims to fill this gap, both in terms of research and application, through its approach that synthesizes three different axes of the field and brings together methods previously addressed separately in the literature within a single workflow.

5. Literature Comparison and Review

Work(Ref)	Used Technology	Focus Area	Patient Related Data?	Main Restrictions and Limitations
García-Robles et al. [1]	Immersive VR/AR	Anatomy Education	**No** (General Model)	Doesn't contain models for an individual.
Liu et al. [6]	Deep learning via CNN	Tumor Detection	**Yes** (Dataset)	Only has the segmentation feature but lacks implementation into reality.
Panesar et al. [11]	3D Print	Surgical Pre-plan	**Yes**	Static Model; can't be readjusted after operation(s).
Crispi et al. [12]	Mixed Reality	Inform Patient	**Yes**	The automated segmentation workflow is not fully integrated; there is a manuel process.
Proposed Project	DL + VR/MR	Education & Surgical Pre-plan	**Yes** (Automatic)	Fulfills every limited aspect with a reliable base.

6. Dataset Description(2nd Report)

6.1 Use of the BraTS 2020 Dataset in the MedVisVR Project

General Definition and Purpose

The BraTS 2020 [21] dataset is an authoritative resource considered the "gold standard" in the field for the automated analysis and segmentation of brain tumors within multimodal Magnetic Resonance Imaging (MRI) scans.

Within the MedVisVR project, this dataset is utilized for the precise detection of glioma-derived pathologies, including High-Grade Gliomas and Low-Grade Gliomas. The primary objective of the project is to transform the high-accuracy segmentation data derived from this dataset into clinically meaningful, 3D interactive models within a Virtual Reality environment.

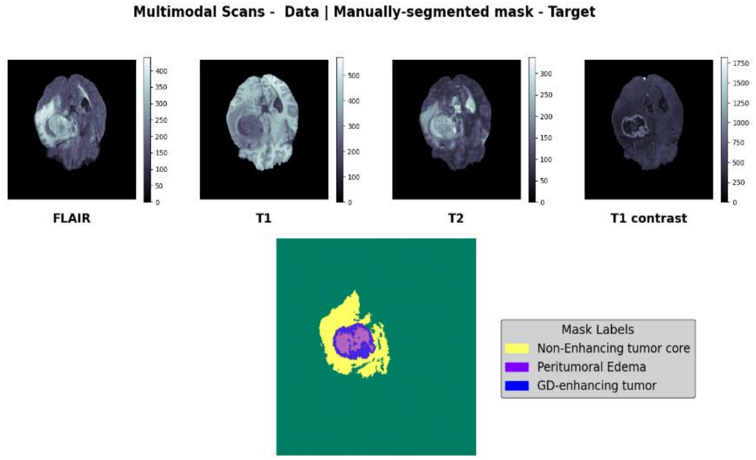
Purpose of Use and Segmentation Objectives

Within the scope of the project, BraTS 2020 data is processed to achieve the automated decomposition of heterogeneous glioma sub-components across multimodal MRI sequences.

The resulting segmentation outputs define the anatomical boundaries, volume, and internal structure of the tumor, forming the foundation for the 3D pathological objects generated on the MedVisVR platform. This process allows clinicians to examine the tumor structure in depth and through multiple layers.

Imaging Data Characteristics

The dataset consists of pre-operative MRI scans collected from 19 different institutions, utilizing various clinical protocols and diverse MRI scanners.



1. **Modalities** Each patient case includes four co-registered MRI sequences:
   - **T1:** Native T1-weighted.
   - **T1ce**: Contrast-enhanced T1-weighted (Gadolinium - T1Gd).
   - **T2:** T2-weighted.
   - **FLAIR:** T2-weighted, Fluid-Attenuated Inversion Recovery.

2. **Format and Pre-processing** : To ensure model stability and data consistency, the raw data has been standardized through the following procedures:

   - **Co-registration:** All modalities have been aligned to the same anatomical space.
   -**Resolution:** Images have been resampled to a 1 mm³ isotropic.

-Skull-stripping: Non-brain tissues and the skull have been removed to focus exclusively on intracranial structures.

-File Format: The original data is provided in NIfTI (.nii.gz) format.

-Optimization: Within the MedVisVR architecture, volume data is converted and utilized in HDF5 (.h5) format to increase memory efficiency during training and visualization processes.

3. **Data Volume**

-**Training Set:** 369 cases.

-**Validation Set:** 125 cases.

MedVisVR Segmentation and Visualization Approach

The standard labels in the BraTS dataset have been restructured within MedVisVR to provide clinicians with the clearest possible imagery in a VR environment:

-Tumor Core: The necrotic/non-enhancing regions and the active enhancing tumor tissue are merged and processed as a single structure. In the VR environment, this is visualized as a solid, holistic object representing the primary mass of the tumor.

-Peritumoral Edema: The edema tissue surrounding the tumor is maintained as a separate layer. During visualization, it is presented with distinct color coding or transparency settings to demonstrate the spread and impact of the tumor on healthy tissue.

This approach ensures that the relationship between the tumor core and its infiltration into surrounding tissues can be intuitively distinguished during surgical planning or educational simulations.

6.2 Use of the ISLES 2022 Dataset in the MedVisVR Project

General Definition and Purpose

The ISLES 2022 [2] (Ischemic Stroke Lesion Segmentation Challenge) dataset is a reliable reference dataset featuring multi-institutional, expert-level annotations focused on lesion segmentation in acute ischemic stroke cases. In the MedVisVR project, this dataset was utilized for the accurate detection and volumetric analysis of ischemic lesions (infarct core), as well as their evaluation in 3D within the context of brain anatomy in a VR environment.

Imaging Data Characteristics

The ISLES 2022 dataset includes three fundamental MR sequences for each case, accompanied by expert-generated ischemic lesion annotations: FLAIR, Diffusion-Weighted Imaging (DWI), and corresponding ADC maps. These modalities provide critical clinical information, particularly in defining the infarct core (dead tissue) and clarifying lesion boundaries during the acute stroke process.

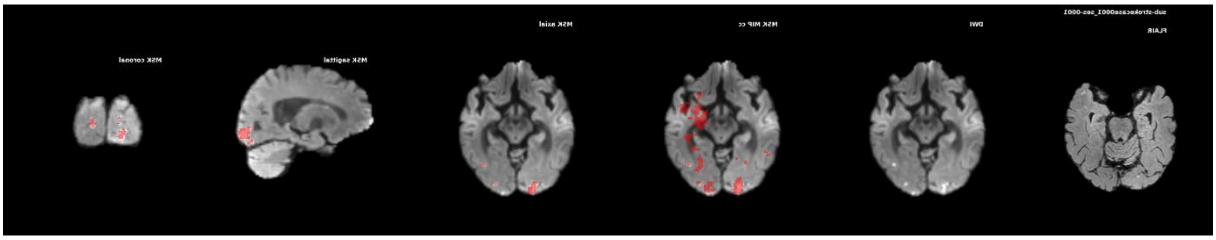


Figure 1 ISLES 2022

Data Format and Pre-processing

All images and annotations were provided in NIfTI format in compliance with the **BIDS (Brain Imaging Data Structure)** standard, which is widely used in brain imaging studies. Images were shared in the patients' native space, and no additional spatial alignment was applied. To protect patient privacy, skull-stripping was performed on all data. For applicable cases, DICOM metadata was provided in JSON format to ensure the traceability of imaging parameters.

Data Source and Imaging Centers

The dataset was collected retrospectively from three different stroke centers using MR systems with varying magnetic field strengths. This diversity allowed for the evaluation of the robustness of the methods developed within MedVisVR against different devices and clinical scenarios.

Purpose of Use and Segmentation Objectives

Annotation Structure

Within the scope of ISLES 2022, all cases were labeled under a single class: ischemic lesion (infarct core). These annotations were created by expert clinicians, providing a reliable foundation for segmentation accuracy.

Data Volume

The ISLES 2022 dataset used in the project consists of approximately 250 expert-annotated acute ischemic stroke cases. This data served as the primary input for training the lesion segmentation model and developing the MedVisVR visualization components.

MedVisVR Segmentation and Visualization Approach

Implementation Approach in MedVisVR

In the MedVisVR project, ISLES 2022 data was integrated for the precise segmentation of ischemic lesions using diffusion-based MR images. The resulting segmentation results were converted into 3D models to reveal the spatial location and volume of the lesion within the brain anatomy. In the VR environment, ischemic lesions are represented with a blue color code to facilitate visual differentiation.

Through this method, the aim is to enable clinicians to evaluate the spread, boundaries, and relationship of the stroke lesion with brain tissue in a more intuitive and holistic way.

6.3 Use of the TotalSegmentator Dataset in the MedVisVR Project

General Characteristics of the Dataset

TotalSegmentator dataset [23] [24] is a comprehensive anatomical segmentation dataset created from large-scale, whole-body computed tomography (CT) scans. Within MedVisVR, it provided an infrastructure that strengthens the relationship between neurological pathologies and surrounding tissues, specifically by taking the head and upper body anatomy as a reference.

Imaging Data Characteristics

The dataset covers versions v1 and v2 published via the Zenodo platform. The first version includes 1228 CT scans, while version v2 has expanded data diversity and annotation scope. All data is provided in NIfTI format, making it compatible with MedVisVR's image processing and visualization components.

Purpose of Use and Segmentation Objectives

The TotalSegmentator dataset includes detailed labels covering 104 different anatomical structures, including bone structures, organs, vessels, and soft tissues. This rich annotation structure enabled the alignment of pathological segmentations with anatomical references in MedVisVR, presenting them in a more clinically meaningful context.

MedVisVR Segmentation and Visualization Approach

In the MedVisVR project, TotalSegmentator data was used independently of pathology-focused segmentations (e.g., tumors or ischemic lesions) to create an anatomical framework. Thanks to this framework, the pathological structures shown in the VR environment were placed in the correct position and scale within a realistic head anatomy.

This approach allows users to intuitively evaluate not only the pathology itself but also its relationship with surrounding anatomical structures in 3D.

6.4 Use of the CHAOS – Combined (CT-MR) Healthy Abdominal Organ Dataset in the MedVisVR Project

General Definition and Purpose

The CHAOS -Combined (CT-MR) Healthy Abdominal Organ [25] dataset is a comprehensive reference dataset prepared for abdominal organ segmentation, containing both CT and MR images. The dataset consists of images obtained from healthy individuals, allowing for the detailed and accurate modeling of abdominal anatomy. In this regard, it provides a reliable basis for determining organ boundaries, volumetric analysis, and method evaluation in clinical applications.

Purpose of Use and Segmentation Objectives

In the MedVisVR project, the CHAOS dataset was not used directly for pathological segmentation; instead, it was evaluated as a supporting resource for creating anatomical context and visually defining organ boundaries. Specifically, referencing the torso anatomy made it possible to present pathological structures in a more meaningful and realistic context within the VR environment.

Imaging Data Characteristics

The CHAOS dataset includes abdominal CT and MR images obtained from different individuals. Both imaging modalities focus on normal organ structures, i.e., those containing no pathological findings. This allows for the clear representation of the anatomical boundaries of primary abdominal organs such as the liver, kidneys, and spleen.

Format and Pre-processing

While the dataset was originally provided in clinical imaging formats, the images were converted to NIfTI (.nii.gz) format within the MedVisVR project to align with standard image processing and visualization workflows. The data structure was organized with a training and testing split; the training data includes both the images and their corresponding manual segmentation masks.

Organ Labels

Labeling in the CHAOS dataset covers different organs depending on the modality. While CT images primarily feature liver segmentation, MR images contain segmentations of multiple abdominal organs, including the liver as well as the right and left kidneys and the spleen. This structure allows for the modeling of anatomical relationships between different organs.

MedVisVR Segmentation and Visualization Approach

In the MedVisVR project, healthy organ segmentations obtained from the CHAOS dataset were used to create 3D anatomical models. These models were integrated into the VR environment to show the spatial relationship of pathological segmentations with surrounding organs. Thus, pathological structures were positioned within a realistic anatomical environment rather than as abstract objects.

This approach allows users to evaluate not only the pathology itself but also its relationship with surrounding organs in a holistic manner. In this way, the CHAOS dataset has contributed to the project as an important component that strengthens the anatomical consistency of MedVisVR and enriches the visualization experience.

6.5 CT-ORG (CT Volumes with Multiple Organ Segmentations)

General Definition and Purpose

The CT-ORG [26] dataset, used to ensure that the MedVisVR project masters the entirety of human anatomy rather than just specific pathologies (tumors, hemorrhages, etc.), is the fundamental building block that enhances the model's "scene perception." Compiled by The Cancer Imaging Archive (TCIA), this collection is used to enable our AI model to distinguish multiple organ structures in different body regions simultaneously and with high accuracy (Multi-Organ Segmentation).

Purpose of Use and Segmentation Objectives

In real-world scenarios, medical images are not standardized; patient posture, weight, or organ placement can vary. CT-ORG plays a critical role in training our model against these variations.

The primary function of this dataset in the project is:

- **Contextual Learning:** It enables the model to not only find the "liver" but to understand the liver's proximity to the lungs and bone structures. Consequently, organs are visualized in the VR environment not as disjointed parts, but as an anatomical whole (like a completed jigsaw puzzle).

Imaging Data Characteristics

Instead of data from a single clinical study, CT-ORG consists of 140 CT series compiled from various sources (e.g., the Liver Tumor Segmentation Challenge - LiTS, etc.).

This "mixed" structure is a major advantage for our project:

- **Generalizability:** Since the data comes from different CT devices and centers, our model does not "overfit" to the image quality or noise type of a single hospital. Instead, it gains a robust structure capable of recognizing organ boundaries even under varying imaging conditions.

Targeted Anatomical Structures

The dataset includes segmentation masks for the following radiologically critical organs covering a large portion of the body (from the chest to the pelvis):

1. **Lungs:** To define the boundaries of the thoracic region and create a respiratory system reference.
2. **Liver:** As the most dominant organ of the abdomen, serving as the central point of abdominal navigation.
3. **Kidneys:** Critical references for modeling the retroperitoneal (behind the abdominal lining) region.
4. **Bladder:** Defines the lower boundary of the pelvic region.

5. **Bones:** Forming the skeletal frame in which all these soft tissues are placed, providing the basic reference for the user's orientation in the VR environment.
6. **Brain**: Available in some series, used to ensure integrity in head region scans.

MedVisVR Segmentation and Visualization Approach

In their raw form, CT-ORG data have different slice thicknesses and resolutions. Before being fed into the MedVisVR "Anatomical Context Engine," they underwent the following processes:

- **Hounsfield Unit (HU) Windowing: s**Density ranges where different organs (soft tissue vs. bone) are most clearly visible were normalized. This allows the model to focus on tissue characteristics independently of the image's brightness/contrast settings.
- **Format Uniformity:** All volumes and masks were converted to the project standard, NIfTI (.nii.gz) format.

Quality Control: Since the masks in the dataset were produced with automated methods and corrected by experts (hybrid annotation), they underwent a final check for anatomical consistency before training.

6.6 VerSe'20

General Definition and Purpose

The **VerSe'20** [27] dataset is one of the primary reference datasets ensuring that pathological structures are placed within a realistic anatomical framework in the MedVisVR system. Specifically, in terms of modeling the head and upper spine anatomy, this dataset, which focuses on spinal segmentation and vertebrae labeling tasks was used to enhance the accuracy of the anatomical context.

Purpose of Use and Segmentation Objectives

VerSe'20 is a large-scale dataset created from CT images with the goals of:

- Automated segmentation of the vertebrae forming the spine.
- Accurate labeling of the anatomical level of each vertebra.

In the context of MedVisVR, this dataset enables pathological segmentations to be presented within a clinically meaningful skeletal and head anatomy, rather than as isolated objects in space.

Imaging Data Characteristics

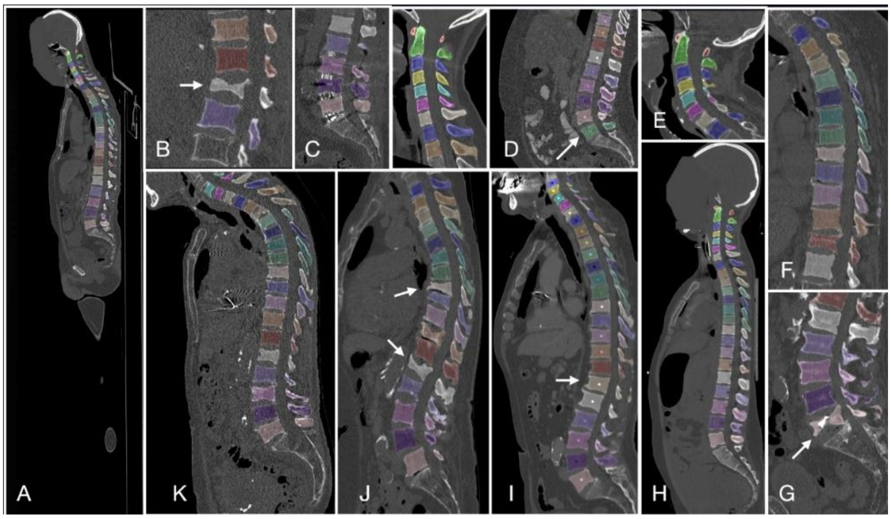


Figure 2 VerSe'20

- **Modality:**
  Computed Tomography (CT)
- **Format:**
  All images and segmentation masks are provided in **NIfTI (.nii / .nii.gz)** format.

This format is fully compatible with the MedVisVR pre-processing pipeline, allowing for the direct transfer of 3D volumes into the VR environment.

Annotations and Labeling Structure

For each case, the VerSe'20 dataset includes the following components:

- Voxel-based vertebrae segmentation masks, allowing for the 3D decomposition of each vertebra in the spine.
- Anatomical level labels defining the positions of the vertebrae in the cervical, thoracic, and lumbar regions.

MedVisVR Segmentation and Visualization Approach

Through this structure, spinal anatomy can be modeled semantically as well as geometrically.

This anatomical structure is integrated through a fusion process with:

Ischemic lesions derived from ISLES 2022.

Consequently, the spatial relationship between these pathologies and the cranial and spinal anatomy is presented with high fidelity within the VR environment. This approach ensures that the pathological findings are not viewed in isolation but are contextualized within the patient's actual anatomical framework.

Systemic Contribution

The primary contributions of VerSe'20 to the MedVisVR architecture are as follows:

- Preventing the abstract visualization of pathological structures without anatomical references.
- Reducing errors in scale, position, and orientation within the VR environment.
- Enabling clinical users to evaluate the pathology in relation to the skeletal and head anatomy.

In this way, MedVisVR evolves from a system focused solely on segmentation accuracy into a context-aware visualization platform for clinical decision support.

7. Dataset Tables

7.1 Primary Research Datasets

Dataset	Imaging Modality	Target Structures (Labels)	Primary Role in MedVisVR	Data Format
BraTS 2020	**Multimodal MRI** (T1, T1ce, T2, FLAIR)	**Glioma Tumor Sub-regions:** • Tumor Core (NCR/NET + ET) •Peritumoral Edema	**Pathological Visualization:** Detection and segmentation of brain tumors and their presentation as 3D pathological objects within the VR environment.	NIfTI (Converted to HDF5)
ISLES 2022	**MRI** (DWI, ADC, FLAIR)	**Ischemic Lesion:** • Infarct Core (Dead Tissue)	**Pathological Visualization:** Delineation of lesion boundaries in acute stroke cases and volumetric representation using a blue color code.	NIfTI (BIDS Standard)
TotalSegmentator (v1 & v2)	**CT**	**Anatomical Structures:** • Skull • Vertebrae • Skin	**Anatomical Context:** Creating a reference skeleton to position pathologies within a realistic head/body structure rather than in empty space.	NIfTI
CHAOS	**CT & MRI**	**Abdominal Organs:**	**Anatomical Context:**	NIfTI

Dataset	Imaging Modality	Target Structures (Labels)	Primary Role in MedVisVR	Data Format
		• Liver • Kidneys • Spleen	Establishing spatial relationships between pathologies and surrounding tissues by referencing healthy organ boundaries.	
CT-ORG	**CT**	**Multi-Organs:** • Lungs, Liver, Bone Structure, Bladder, Brain	**Scene Perception & Generalizability:** Enabling the model to learn "anatomical integrity" (scene perception) against variations in patient posture and device differences.	NIfTI
VerSe'20	**CT**	**Spine:** • Vertebra Segmentation •Anatomical Level Labels	**Anatomical Context:** Ensuring proper orientation, particularly in the head and neck region, and modeling the spinal anatomy.	NIfTI

7.2 External Validation & Reference Datasets

Dataset	Imaging Modality	Target Structures (Labels)	Primary Role / Intended Use	Data Format
MSD (Medical Segmentation Decathlon)	**Multimodal** **(CT, MRI - T1, T2, FLAIR, ADC)**	**10 Different Organs & Pathologies:** • Brain (Glioma) • Heart (Atrium) • Liver & Tumor •Hippocampus • Prostate • Lung • Pancreas • Hepatic Vessel • Spleen • Colon	**Generalizability Testing:** Developing and comparing "general-purpose" segmentation models capable of working across different anatomical structures and modalities.	**NIfTI** (With standardized directory structure and JSON metadata)
TCIA – Glioma, Head-Neck, Lung	**Multimodal** (MRI, CT, PET, Pathology)	**Cancer-Focused Structures:** • Glioma: Tumor sub-regions (Necrosis, Edema) • Head-Neck: Tumor volume and organs at risk • Lung: Nodules and masses	**Clinical Research & Radiomics:** Treatment response assessment in real clinical scenarios, radiotherapy planning, and genomic data matching.	**DICOM** (Raw clinical image format, requires processing)
Brain MRI Dataset (Multiple Sclerosis)	**MRI** (T1, T2, FLAIR)	**Pathological Lesions:** • MS (Multiple Sclerosis) Plaques • White Matter Lesions	**Disease Monitoring & Training:** Model training for detection and volumetric analysis of MS lesions, and correlation with patient disability status (EDSS).	**NIfTI** (Usually provided in processed or raw format)

8. Preprocessing And Standardization Pipeline

8.1 Executive Summary

This report describes the data preprocessing pipeline designed for the BraTS 2020 dataset, which contains multi-modal MRI scans. The goal of this pipeline is to prepare the data for training 3D Convolutional Neural Networks specifically 3D U-Net architecture. The workflow is optimized to handle hardware limitations, reduce data heterogeneity, and mitigate class imbalance issues inherent in tumor segmentation tasks.

8.2. Technical Specifications (Input/Output Overview)

Parameter	Raw Data (Input)	Processed Data (Output)
Dimensions (Tensor Shape)	(240, 240, 155)	(128, 128, 128) (Patch-Based)
Channel Structure	4 Separate Files (T1, T1ce, T2, FLAIR)	(4, 128, 128, 128) (Stacked Tensor)
Labels	0, 1, 2, 4 (Original Classes)	3-Channel Multi-Label (WT, TC, ET)
Voxel Resolution	Variable (may be anisotropic)	1.0 × 1.0 × 1.0 mm (Isotropic)
Orientation	Inconsistent	RAS (Right–Anterior–Superior)
Pixel Intensities	Raw MRI Intensity (0–3000+)	Normalized (Mean = 0, Std = 1)

8.3 Processing Pipeline

The preprocessing architecture is implemented using the MONAI library and follows this sequential workflow:

1. **Data Loading**
   Loading and stacking the four MRI modalities.
2. **Label Mapping**
   Converting tumor labels into clinical subregions (WT / TC / ET).
3. **Spatial Transforms**
   - Orientationd(RAS): Standardizes image orientation.
   - Spacingd(1mm): Resamples all scans to 1 mm isotropic spacing.

4. **Intensity Normalization**
   Z-score normalization applied only to brain tissue.
5. **Class-Balanced Patch Extraction**
   Using RandCropByPosNegLabeld to ensure patches with tumor regions are preferentially sampled.
6. **Data Augmentation**
   Random flips and rotations to increase data diversity.
7. **Export**
   Writing the processed tensors back to .nii files.

8.4 Methodology Details

8.4.1. Multi-Modality and Semantic Transformation

A multi-modal learning strategy is used to help the model better capture tumor boundaries. Original labels are reorganized into three clinically relevant regions:

- **Whole Tumor (WT)**
- **Tumor Core (TC)**
- **Enhancing Tumor (ET)**

This multi-label representation is known to improve segmentation performance, especially Dice scores.

8.4.2. Computational Strategy: Label-Guided Sampling

The raw MRI volumes (240 × 240 × 155) are large and memory-intensive. Additionally, tumor regions occupy only a small fraction of the volume, making naïve random cropping ineffective most patches contain only background brain tissue.

To address this, the pipeline uses **RandCropByPosNegLabeld**, which ensures:

- Training volumes are cropped into 128 × 128 × 128 patches.
- Each batch contains patches in a **2:1 ratio of tumor-containing vs. healthy tissue**.
- Tumor regions are consistently included during training, significantly reducing class imbalance.

8.5 Error Handling and Quality Control

To ensure robustness, the following safeguards are implemented:

- Missing File Detection: The pipeline verifies file integrity before processing.
- Channel Consistency: EnsureChannelFirstd enforces PyTorch’s (C, H, W, D) channel ordering.
- Determinism: set_determinism (seed=0) ensures reproducible augmentation and transformations.

8.6 Conclusion

The revised preprocessing pipeline produces a dataset that is not only technically standardized but also optimized for high-performance tumor segmentation. Through normalization, orientation correction, label refinement, and class-balanced sampling, the pipeline effectively prepares MRI data for training 3D U-Net models and enhances overall Dice score performance.

9. Ethical Considerations and Data Privacy

9.1. Use of Open-Access Research Repositories

The MedVisVR project is built upon established, open-source medical datasets (BraTS 2020, ISLES 2022, TotalSegmentator, CT-ORG, VerSe’20, and CHAOS). These datasets are hosted by globally recognized academic and clinical repositories, such as The Cancer Imaging Archive (TCIA) and Zenodo. By utilizing these pre-existing, publicly available resources, the project ensures that:

Ethical Oversight: All data was originally collected under the approval of the respective Institutional Review Boards (IRBs) of the contributing hospitals.

Conflict Mitigation: Using open-access data eliminates ethical conflicts regarding primary data collection and patient consent, as the data has already been cleared for international research and development purposes.

9.2. Adherence to Licensing and Usage Policies

The project strictly complies with the specific licensing terms of each dataset (e.g., Creative Commons Attribution 4.0 International - CC BY 4.0). All data is used solely for the stated research, development, and educational goals of MedVisVR. In accordance with competition rules (such as those of the BraTS and ISLES challenges), the project ensures that any comparative results or findings are reported transparently and ethically.

9.3 Scientific Integrity and Secondary Data Use

As a secondary data user, the MedVisVR team maintains scientific integrity by referencing all original authors and institutions. The reliance on these "gold standard" benchmarks ensures that the project's results are reproducible and comparable with other state-of-the-art methods in the medical imaging community.

10.Methodology(Part 3)

Abstract

This report is intended to present a comprehensive framework for the development of a Virtual Reality (VR) based brain tumor visualization system. Advanced deep learning segmentation techniques are integrated with immersive VR technology to create an interactive 3D visualization platform for brain tumor analysis and surgical pre-planning. Multi-modal MRI data from the BraTS2020 dataset is processed, automated tumor segmentation is performed using the nnU-Net architecture, and the results are rendered in an immersive VR environment using the Meta Quest 3. This approach is aimed at enhancing medical education, treatment planning, and patient communication through the intuitive 3D visualization of complex neurological structures.

11.1 Background

Brain tumors represent one of the most challenging conditions in modern medicine, requiring precise diagnosis and treatment planning. Traditional 2D medical imaging techniques, while valuable, often fail to convey the complex three-dimensional spatial relationships between tumors and surrounding brain structures. The advent of Virtual Reality (VR) technology presents a transformative opportunity to visualize and interact with medical imaging data in ways that closely mirror the actual anatomical reality.

The BraTS (Brain Tumor Segmentation) Challenge has established itself as the premier benchmark for evaluating brain tumor segmentation algorithms. The BraTS2020 dataset provides multi-parametric MRI scans including T1-weighted, T1-weighted with contrast enhancement (T1ce), T2-weighted, and FLAIR sequences, along with expert-annotated tumor segmentation masks distinguishing between different tumor sub-regions.

11.2 Project Objectives

This project aims to develop an end-to-end pipeline that:

• Automatically segments brain tumors from multi-modal MRI scans using state-of-the-art deep learning techniques

• Converts medical imaging data into formats compatible with real-time 3D rendering engines

• Renders the composite brain model in an immersive VR environment using Meta Quest 3

• Provides intuitive interaction mechanisms for exploring and analyzing tumor characteristics in VR

11.3 Significance

The proposed system offers several significant advantages over traditional visualization methods:

Enhanced Spatial Understanding: VR enables clinicians to perceive tumor location, size, and relationship to critical brain structures in true 3D space, improving surgical planning and risk assessment.

Improved Patient Communication: Patients and families can better understand their condition through intuitive, immersive visualization rather than abstract 2D scans.Medical Education: Medical students and residents can explore realistic tumor cases in a risk-free, interactive environment.

Reproducibility and Automation: The automated segmentation pipeline ensures consistent, reproducible results across different cases and operators

12. Part I: Brain Tumor Segmentation and Processing

12.1 Dataset: BraTS2020

The BraTS2020 dataset serves as the foundation for our tumor segmentation component. This dataset contains multi-institutional, multi-parametric MRI scans of glioblastoma and lower-grade glioma patients.

Dataset Characteristics:

• Imaging Modalities: Four MRI sequences per patient (T1, T1ce, T2, FLAIR)

• Volume Dimensions: Typically 240×240×155 voxels

• Voxel Resolution: $1mm^3$ isotropic resolution

• Annotation Labels: 0 (background), 1 (necrotic/non-enhancing tumor core), 2 (peritumoral edema), 4 (enhancing tumor)

• Pre-processing: Skull-stripped, co-registered to the same anatomical template, and interpolated to the same resolution

12.2 Tumor Segmentation: nnU-Net Architecture

The nnU-Net (no-new-Net) deep learning framework is employed for automated tumor segmentation. This architecture is widely recognized as the state-of-the-art standard in biomedical image segmentation, having consistently demonstrated superior performance and robustness across diverse datasets, including the BraTS benchmarks.

12.2.1 Architecture Overview

nnU-Net is a self-configuring semantic segmentation method that automatically adapts to new datasets. The architecture is based on the U-Net encoder-decoder structure with the following key components:

• **Encoder Path:** Progressive downsampling through convolutional blocks to capture semantic features at multiple scales

• **Decoder Path:** Symmetric upsampling path with skip connections to recover spatial resolution

• **Skip Connections:** Direct connections between encoder and decoder at matching resolutions to preserve fine-grained spatial information

• **Deep Supervision:** Auxiliary outputs at multiple decoder levels to improve gradient flow during training

12.2.2 Input Processing

The model processes all four MRI modalities simultaneously as a 4-channel input volume. Pre-processing steps include:

Data Preparation:

• Loading of all four MRI sequences (FLAIR, T1, T1ce, T2) from the BraTS validation set

• **Intensity normalization:** Z-score normalization per modality to standardize intensity distributions

• **Resampling:** Ensuring consistent voxel spacing across all modalities

• **Padding/Cropping:** Adjusting volumes to network-compatible dimensions

12.2.3 Inference and Output

The inference process is executed using the following command structure:

```
nnUNetv2_predict -i imageTsv/ -o predictions/ -d 500 -c 3d_fullres -f 0
```

where parameters specify input directory (-i), output directory (-o), dataset ID (-d), configuration (-c), and fold (-f).

Output Characteristics:

• Format: 3D segmentation masks in NIfTI format

• Labels: Multi-class segmentation distinguishing necrotic core (Label 1), edema (Label 2), and enhancing tumor (Label 4)

• Dimensions: Matches input dimensions (240×240×155 voxels)

• Preservation: Maintains tumor morphology and regional structure as captured in the original patient scan

12.3 Brain Tissue Detection in Prototype

To integrate the segmented tumor into a standardized visualization environment, a brain prototype (dummy brain model) is utilized. This prototype serves as the anatomical reference into which patient-specific tumor data is integrated.

12.3.1 Prototype Characteristics

The brain prototype is a standardized anatomical model representing average brain structure. Key properties include:

• Anatomical Accuracy: Based on averaged MRI data from subjects

• Standard Space: Aligned to a common coordinate system for consistent spatial referencing

• Tissue Segmentation: Pre-segmented into gray matter, white matter, and cerebrospinal fluid regions

• File Format: NIfTI (.nii) format for compatibility with medical imaging tools

12.3.2 Brain Tissue Extraction Method

We employ Otsu thresholding combined with morphological operations to extract brain tissue from the prototype. This classical computer vision approach provides robust tissue detection without requiring deep learning models.

Algorithm Steps:

1. Image Loading: Load the brain prototype in NIfTI format

2. Intensity Normalization: Standardize intensity values to a common range

3. Otsu Thresholding: Automatically determine optimal threshold to separate brain tissue from background based on intensity histogram bimodality

4. Morphological Operations: Apply erosion and dilation to remove noise and fill small gaps

5. Connected Component Analysis: Identify and retain the largest connected component as brain tissue

6. Binary Mask Generation: Create a binary mask delineating brain tissue regions

12.4 Spatial Alignment and Integration

A critical step in creating realistic visualizations is accurately aligning the patient-specific tumor mask with the brain prototype. This process ensures that the tumor appears in anatomically plausible locations within the prototype brain.

12.4.1 Spatial Registration Methodology

We implement a rigid transformation approach that preserves tumor morphology while repositioning it within the prototype brain. This method balances anatomical accuracy with computational efficiency.

Method: Center-of-Mass Alignment

The alignment process is based on computing and matching the centers of mass (CoM) of both the tumor and the brain prototype:

Step 1 - Compute Brain Center:

```
brain_center = center_of_mass(brain_tissue)
```

This calculates the geometric center of the brain prototype by averaging the coordinates of all brain tissue voxels.

Step 2 - Compute Tumor Center:

```
tumor_center = center_of_mass(tumor_mask)
```

This determines the geometric center of the segmented tumor volume.

Step 3 - Calculate Shift Vector:

```
shift_vector = brain_center - tumor_center
```

The shift vector represents the translation needed to move the tumor from its original position to the center of the prototype brain.

Step 4 - Apply Rigid Transformation:

```
aligned_mask = shift(tumor_mask, shift_vector)
```

The tumor mask is translated using the shift vector, maintaining its original shape and internal structure.

12.4.2 Rationale for Rigid Transformation

The choice of rigid transformation (translation only, no rotation or scaling) is justified by several factors:

Morphology Preservation: Rigid transformations preserve the exact shape and size of the tumor as extracted from the patient scan, maintaining clinical accuracy.

Visualization Priority: For educational and patient communication purposes, preserving tumor morphology is more important than achieving perfect anatomical registration.

Trade-offs: While more sophisticated deformable registration methods (such as Advanced Normalization Tools - ANTs, or SimpleElastix) would provide better anatomical fit, they come with significantly higher computational costs and risk distorting tumor characteristics. These advanced methods would be more appropriate for surgical planning applications where precise anatomical correspondence is critical.

12.5 Positioning Tumor to Brain Tissue

After spatial alignment, the tumor mask is constrained to appear only within brain tissue regions of the prototype. This step ensures anatomical plausibility by preventing tumor voxels from appearing in non-brain areas.

Implementation:

```
mask_constrained = mask_aligned * brain_mask
```

This element-wise multiplication ensures that only voxels that are both part of the tumor and within brain tissue are retained in the final mask.

12.6 Intensity Enhancement for Visualization

To maximize tumor visibility in volume rendering, we apply region-specific intensity enhancement. This technique mimics the appearance of contrast-enhanced MRI where active tumor regions accumulate gadolinium and appear bright.

12.6.1 Multi-Region Enhancement Strategy

Different tumor sub-regions are enhanced with varying intensities to reflect their biological characteristics and improve visual discrimination:

Region	Original Intensity	Enhanced Intensity	Contrast Ratio
Brain Tissue	$\mu \pm \sigma$	$\mu \pm \sigma$	1.0x (baseline)
Necrotic Core (Label 1)	-	$0.1 \times \mu$	0.1x (very dark)
Edema (Label 2)	-	$1.3 \times \mu$	1.3x (slightly bright)
Enhancing Tumor (Label 4)	-	$4.0 \times$ max	4.0x (ultra bright)

12.6.2 Justification

The extreme contrast (4x maximum intensity for enhancing tumor) ensures that tumor regions are clearly distinguishable in volume rendering, mimicking the appearance of contrast-enhanced MRI where gadolinium accumulation creates bright signals in active tumor tissue. This enhancement is critical for effective visualization in VR, where depth perception and spatial relationships must be immediately apparent.

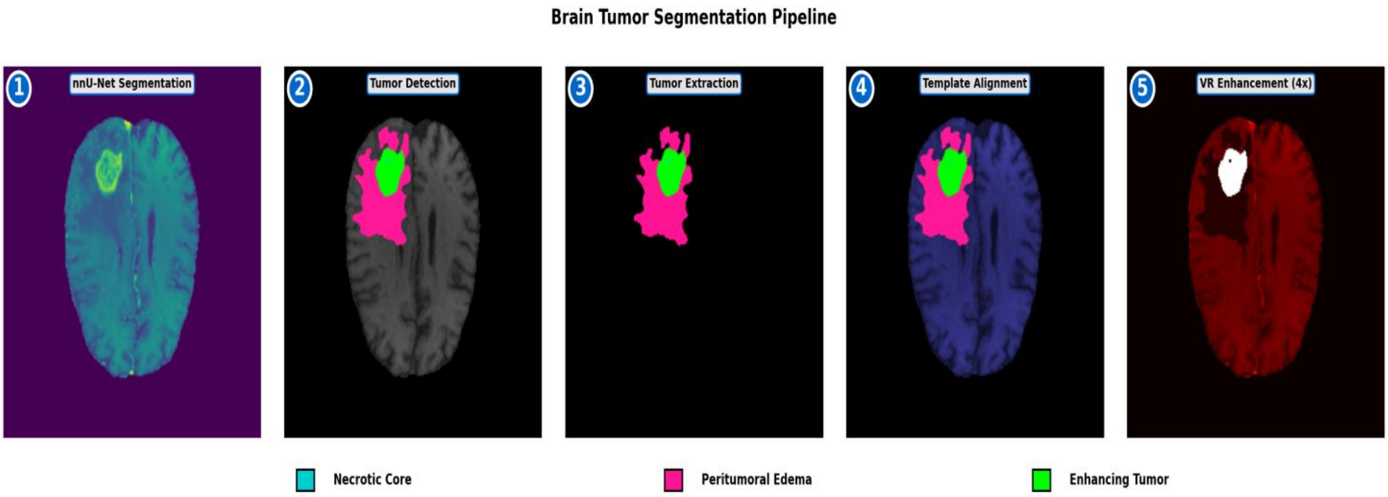
12.7 Multi-Scale Boundary Blending

To create smooth, realistic transitions between tumor and brain tissue, we implement multi-scale Gaussian blending. This approach prevents sharp, artificial-looking boundaries that would detract from visualization quality.

2.7.1 Rationale

This multi-scale approach prevents sharp artificial boundaries while mimicking realistic tumor infiltration patterns. Glioblastoma cells diffusely invade surrounding tissue rather than forming discrete borders, and our blending algorithm recreates this biological characteristic. The varying scales capture both fine-detail infiltration (small σ) and broader transition zones (large σ), producing visually convincing and biologically plausible results.

12.7.2 Figure Example (Segmentation Presentation)



13. Part II: VR Integration and Visualization

The second major component of the project involves converting the processed medical imaging data into a format suitable for real-time 3D rendering and implementing an immersive VR visualization system using Unity and Meta Quest 3.

13.1 Data Format Conversion

Medical imaging data from Part I exists in NIfTI format (.nii), which is not directly compatible with real-time 3D engines. We must convert this volumetric data into formats that Unity can efficiently load and render.

3.1.1 RAW Volume Format

RAW format represents volumetric data as a continuous stream of voxel values without compression or metadata. This simple format is ideal for direct GPU upload and volume rendering.

Conversion Process:

1. Load NIfTI Volume: Read the composite brain-with-tumor volume generated in Part I using nibabel library

2. Normalize Intensity: Scale voxel values to 0-255 range (8-bit unsigned integers) for efficient storage and rendering

3. Extract Voxel Array: Obtain the raw numpy array representing the 3D volume

4. Flatten and Export: Serialize the 3D array to a 1D byte stream and write to .raw file

13.2 Unity Integration

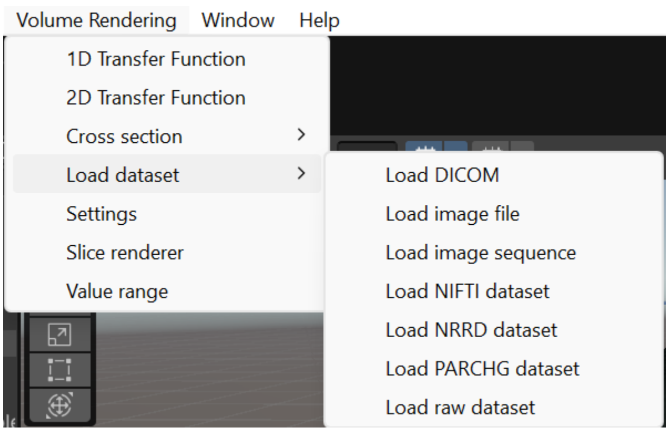
13.2.1 UnityVolumeRendering Plugin

Unity integration for this project is accomplished using the UnityVolumeRendering open-source plugin developed by Matias Lavik (mlavik1). This comprehensive volume rendering solution is specifically designed for medical imaging applications and provides robust support for multiple medical imaging formats including NIfTI (.nii), DICOM, NRRD, and RAW formats.

The UnityVolumeRendering plugin represents a mature, production-ready solution with over 520 stars on GitHub and active maintenance. It implements GPU-accelerated ray marching shaders optimized for real-time volume rendering, making it ideal for AR applications where performance is critical. The plugin has been successfully deployed in various medical visualization projects and educational applications, demonstrating its reliability and effectiveness.

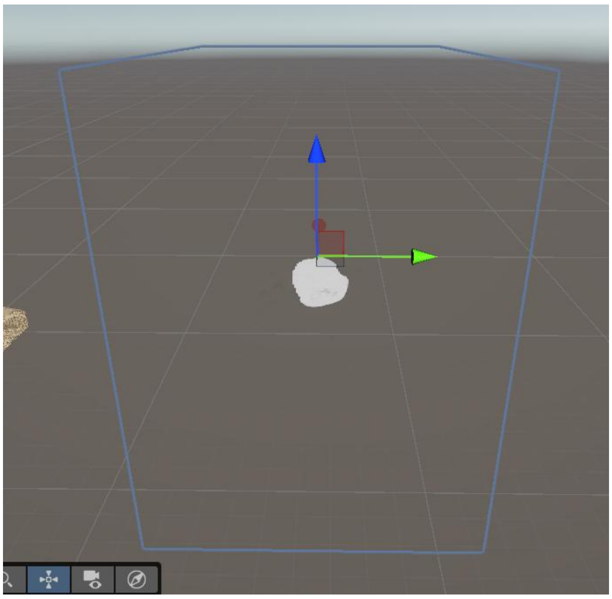
Key Features and Capabilities:

Native NIfTI Support: The plugin includes built-in importers for NIfTI-1 format (.nii and .nii.gz), which is the output format from our processing pipeline. This eliminates the need for manual format conversion and allows direct loading of the composite brain volumes generated in Part I.

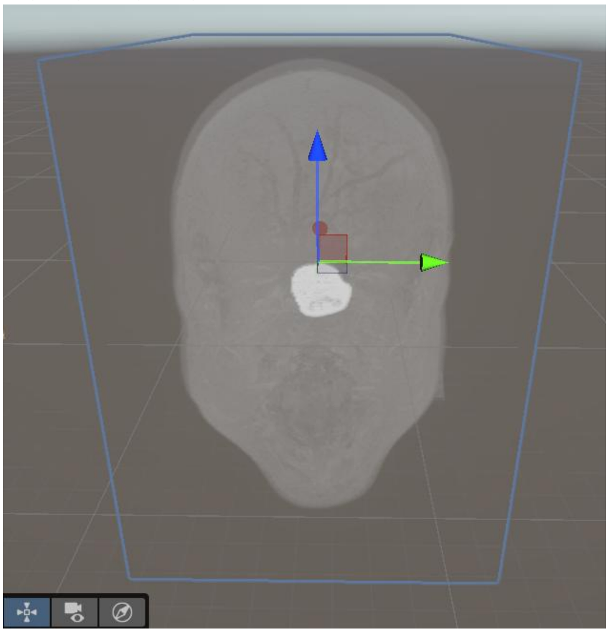


Multiple Rendering Modes: The plugin supports three primary rendering techniques that can be toggled in real-time:

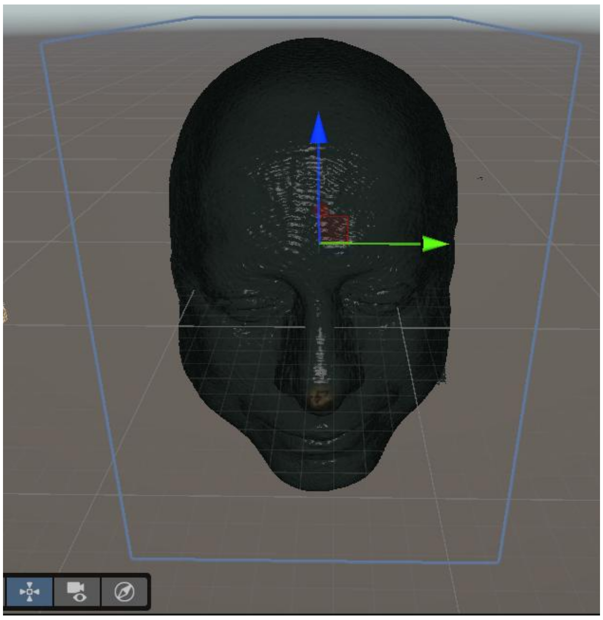
- **Direct Volume Rendering (DVR):** Uses 1D or 2D transfer functions to map voxel intensity to color and opacity, enabling detailed visualization of different tissue types and tumor regions.



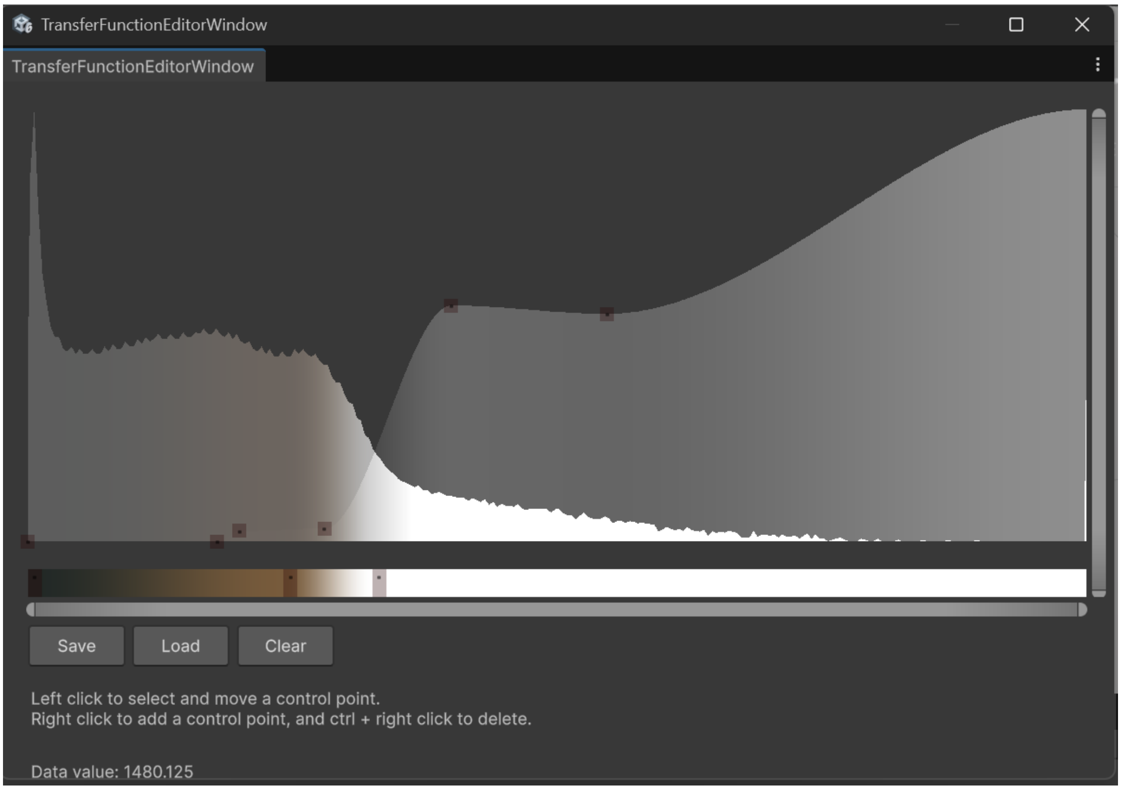
- **Maximum Intensity Projection (MIP):** Projects the maximum intensity along each ray, useful for highlighting bright structures like contrast-enhanced tumors.



- **Isosurface Rendering:** Renders surfaces at specific intensity thresholds, similar to traditional mesh-based rendering but computed directly from volume data.



Transfer Function Editor: Interactive editors for both 1D and 2D transfer functions allow real-time adjustment of visualization parameters. The 1D transfer function maps intensity to color and opacity using an intuitive curve editor, while the 2D transfer function additionally considers gradient magnitude to distinguish surfaces from homogeneous regions. This enables fine-tuned control over how tumor sub-regions appear in the visualization.



VR Compatibility: The plugin is fully compatible with Unity's XR framework and has been tested with various VR headset Meta Quest 3. For Quest 3 deployment, the plugin requires setting the stereo rendering mode to 'multi-pass' in Unity's XR settings to ensure proper stereoscopic rendering.

Advanced Rendering Features:

- Lighting and Shadows: Optional gradient-based lighting calculations provide depth cues and enhance 3D perception. Shadow volumes can be enabled for more realistic rendering, though at higher computational cost.
- Cubic Interpolation: Hardware-accelerated trilinear or tricubic interpolation of the 3D volume texture ensures smooth rendering without visible voxel artifacts.
- Early Ray Termination: Optimization technique that stops ray marching when accumulated opacity exceeds a threshold, significantly improving performance for opaque structures.
- Depth Writing: Proper depth buffer integration allows volume-rendered objects to correctly occlude and be occluded by other 3D geometry in the scene.

Data Pipeline Integration:

Integration Strategy to mitigate the computational overhead associated with parsing raw medical data on a mobile standalone headset, the proposed pipeline decouples data processing from visualization. All data acquisition and manipulation tasks are offloaded to a high-performance PC environment.

PC-Based Preprocessing (Host Environment) Raw NIfTI volumes are imported and processed exclusively within a desktop environment (via the Unity Editor or a dedicated companion application). This stage facilitates comprehensive preprocessing, including the modulation of transfer functions, volumetric cropping, and the refinement of segmentation masks. By leveraging the superior processing capabilities of a desktop workstation, this approach ensures the efficient handling of complex data conversion and scene configuration tasks prior to deployment.

Meta Quest 3 Deployment and Visualization Following optimization on the host PC, the finalized 3D assets are transferred to the Meta Quest 3 hardware. The Head-Mounted Display (HMD) application is architected strictly for the rendering and interactive visualization of these pre-processed volumes. Consequently, the device is relieved of resource-intensive runtime file I/O and NIfTI parsing operations, thereby guaranteeing high-fidelity visualization and stable frame rates during the clinical review process.

13.2.2 Transfer Function Design

Transfer functions map volumetric data intensities to visual properties (color and opacity). Proper transfer function design is critical for effective visualization of different tissue types and tumor regions.

13.2.3 Scene Setup

Unity Scene Components:

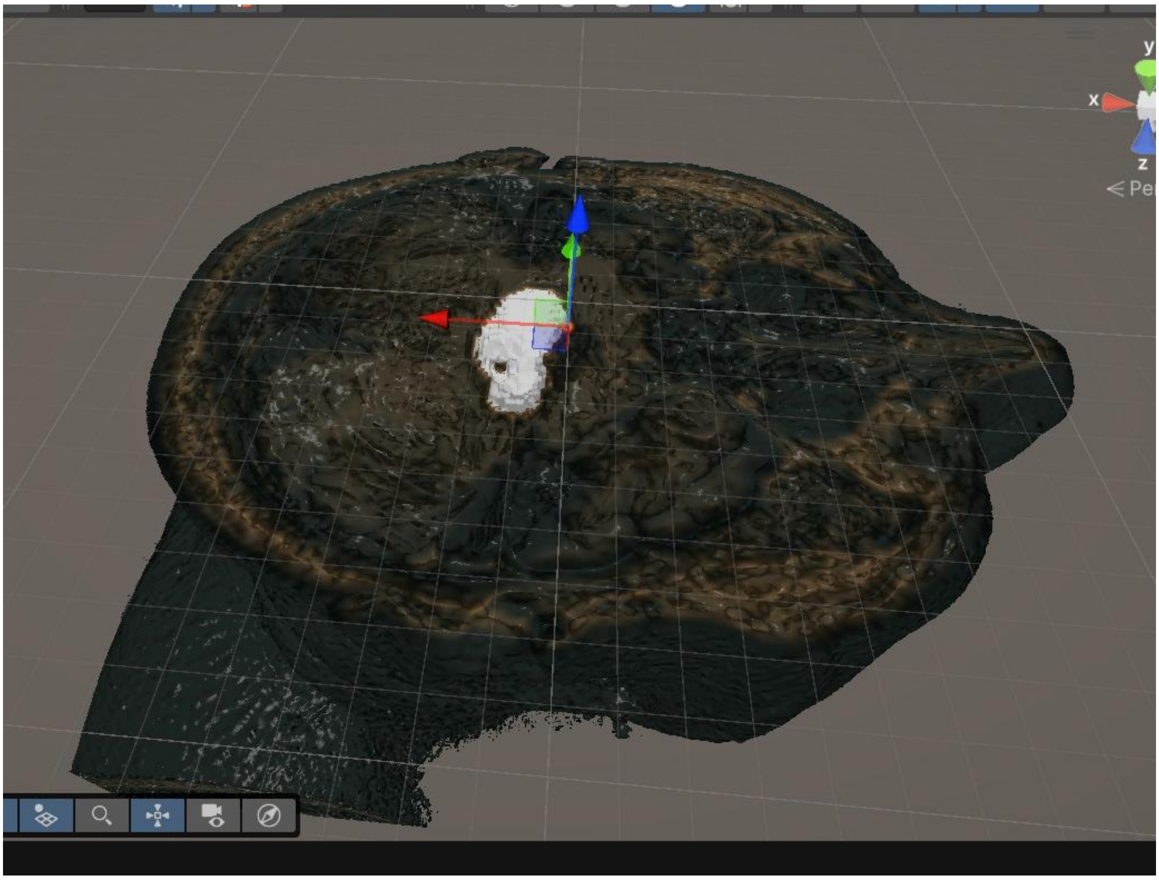
1. VR Camera Rig: XR Interaction Toolkit camera rig configured for Meta Quest 3

2. Volume Object: GameObject containing the loaded brain volume with volume renderer component

3. Lighting: Directional light for surface illumination of UI elements and spatial reference

4. Interaction Components: Ray interactors for hand controllers enabling object manipulation

13.2.4 Figure Example (Scene Image)



13.3 Meta Quest 3 VR Implementation

Meta Quest 3 is a standalone VR headset offering high-resolution displays, inside-out tracking, and powerful mobile processing. Its standalone nature eliminates the need for external computers or sensors, making it ideal for portable medical visualization applications.

13.3.1 Hardware Specifications (Meta Quest 3)

• Display: Dual LCD panels, 2064×2208 pixels per eye, 120Hz refresh rate

• Processor: Qualcomm Snapdragon XR2 Gen 2

• RAM: 8GB LPDDR5

• Tracking: Inside-out 6DOF tracking with 4 wide-angle cameras

• Controllers: Touch Plus controllers with haptic feedback

• Field of View: 110° horizontal, 96° vertical

13.3.2 Development Environment Setup

Required Software and SDKs:

1. Unity 6000.0.6 LTS or later (Long Term Support version for stability)

2. Meta XR SDK: Official Unity package providing Quest-specific features

3. XR Interaction Toolkit: Unity's cross-platform VR framework

4. Android Build Support: Required for deploying to Quest's Android-based OS

5. Meta Quest Developer Hub: For wireless deployment and debugging

Configuration Steps:

1. Install Unity with Android build support

2. Import Meta XR Tools package from Unity Asset Store

3. Configure project settings: Set platform to Android, set minimum API level to Android 10

4. Configure Oculus as active XR runtime

5. Enable developer mode on Quest 3 headset

6. Connect headset via USB or WiFi for deployment

3.3.3 Interaction Design

VR interaction mechanisms must be intuitive and comfortable for extended use. We implement the following interaction paradigms:

Navigation and Manipulation:

• Object Grabbing: Direct hand or controller interaction to grab, rotate, and scale the brain volume

• Scaling: Pinch gesture or button-based scaling to zoom in/out on regions of interest

Visualization Controls:

• Volumetric Slicing: Dynamic cross-sectioning mechanism to visualize internal anatomical structures and pathologies.

• Measurement Tools: Distance measurement between points for quantitative analysis

14. System Architecture

14.1 Overall Pipeline Architecture

The complete system consists of four main processing stages that transform raw MRI data into an immersive VR experience:

Stage 1: Tumor Extraction (nnU-Net)

• Input: BraTS Patient MRI (4 modalities: FLAIR, T1, T1ce, T2)

• Process: nnU-Net 3D Segmentation

• Output: Tumor mask with labels (0=background, 1=necrotic core, 2=edema, 4=enhancing tumor)

Stage 2: Template Preparation

• Input: Brain template (dummy.nii)

• Process: Brain tissue detection using Otsu thresholding and morphological operations

• Output: Binary brain mask delineating tissue regions

Stage 3: Integration

• Inputs: Tumor mask from Stage 1, Brain mask from Stage 2

• Sub-processes:

- Resize tumor mask to match template dimensions
- Align tumor center to brain center (CoM-based rigid registration)
- Constrain tumor to brain tissue regions
- Perform multi-scale Gaussian boundary blending

• Output: Composite brain volume with integrated synthetic tumor

Stage 4: Output Generation

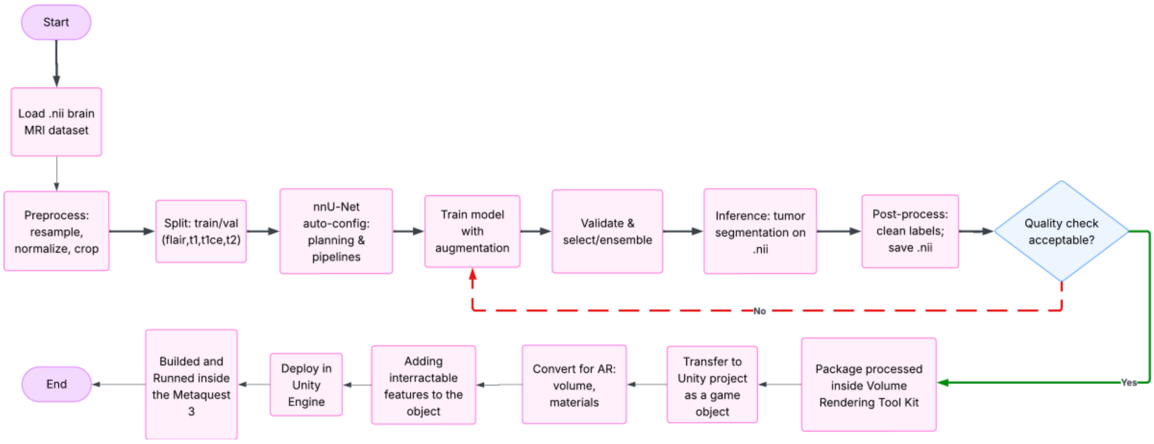
• Input: Composite volume from Stage 3

• Processes:

- Save as NIfTI format for medical imaging applications
- Convert to RAW format for Unity import
- Generate preview images (2D slices)
- Export quality metrics

• Outputs: .nii file, .raw file, metadata.txt, preview images

14.1.1 Pipeline Flowchart



- **Start:** The workflow begins here.
- **Load .nii brain MRI dataset:** The system loads the raw medical brain scans in the NIfTI (.nii) file format.
- **Preprocess:** The images are standardized by resampling, normalizing pixel intensity, and cropping.
- **Split:** The data is divided into training and validation sets using four MRI modalities (FLAIR, T1, T1ce, T2).
- **nnU-Net auto-config:** The nnU-Net framework automatically configures the network architecture and planning based on the dataset.
- **Train model with augmentation:** The AI model is trained using data augmentation (rotating/flipping images) to improve learning.
- **Validate & select/ensemble:** The model is tested for accuracy, and the best version (or a combination of models) is selected.
- **Inference:** The trained AI performs the actual tumor segmentation (prediction) on the .nii files.
- **Post-process:** The output labels are cleaned up to remove noise, and the final segmentation is saved as a .nii file.
- **Quality check acceptable? (Diamond):** If the segmentation is good, proceed to visualization (Green line); if bad, go back and retrain (Red line).
- **Package processed inside Volume Rendering Tool Kit:** The final MRI data is packaged into a format compatible with the volumetric rendering tools.
- **Transfer to Unity project as a game object:** The packaged data is imported into the Unity software as a manipulatable 3D object.

- **Convert for AR:** The object's volume and surface materials are adjusted to look correct in an Augmented Reality environment.
- **Adding interactable features to the object:** Features are added to allow the user to grab, move, or rotate the brain model in 3D space.
- **Deploy in Unity Engine:** The Unity project is prepared and configured for deployment to the hardware.
- **Builded and Runned inside the Metaquest 3:** The application is compiled ("built") and launched on the Meta Quest 3 headset.
- **End:** The process is complete.

14.2 Technology Stack

Processing Pipeline (Python):

• Python 3.9-3.11

• PyTorch 1.12+ (deep learning framework)

• nnU-Net V2 (segmentation framework)

• nibabel (NIfTI file I/O)

• scikit-image (computer vision algorithms)

VR Application (Unity):

• Unity 6000.0 LTS

• C# scripting

• Meta XR Tools

• Meta XR SDK

• Unity Volume Rendering Plugin (mlavik1/UnityVolumeRendering)

Hardware:

• Processing Workstation: NVIDIA GPU (RTX 5060 or higher recommended), 24GB+ RAM

• VR Headset: Meta Quest 3

15. Conclusion

15.1 Summary of Methodology

This methodology report has presented a comprehensive framework for developing a VR-based brain tumor visualization system that bridges advanced medical imaging processing with immersive technology. The proposed system integrates two major components: automated tumor segmentation using state-of-the-art deep learning (Part I) and immersive VR visualization using Meta Quest 3 (Part II).

Part I establishes a robust processing pipeline that transforms multi-modal MRI data from the BraTS2020 dataset into high-quality tumor segmentations using nnU-Net architecture. The segmented tumors are then integrated into a standardized brain template through spatial alignment, intensity enhancement, and multi-scale boundary blending, creating anatomically plausible synthetic volumes suitable for visualization.

Part II details the conversion of medical imaging data to VR-compatible formats and the implementation of an immersive visualization environment in Unity. The system leverages Meta Quest 3's standalone VR capabilities to provide intuitive, interactive 3D exploration of brain tumor anatomy without requiring expensive workstations or tethered hardware.

15.2 Key Innovations

The proposed system introduces several innovative aspects:

1. End-to-End Automation: Complete pipeline from raw MRI scans to VR-ready volumes without manual intervention

2. Ultra-Visible Enhancement: Novel intensity modulation scheme (4x contrast) specifically designed for volume rendering clarity

3. Multi-Scale Blending: Biologically-inspired boundary smoothing that mimics realistic tumor infiltration patterns

4. Standalone VR Deployment: First implementation targeting standalone mobile VR (Meta Quest 3) for medical volume visualization

5. Immersive Pre-operative Assessment: Designed to facilitate complex neurosurgical decision-making, the platform allows for precise visualization of deep brain structures, aiding in the determination of optimal surgical corridors and risk minimization.

6. Interactive Training & Communication: The system adapts its high-resolution rendering capabilities for medical education, providing an intuitive interface for teaching complex neuroanatomy to trainees and explaining surgical procedures to patients.

15.3 Expected Impact

The successful implementation of this system is expected to have significant impacts across multiple domains:

Medical Education: Medical students and residents will gain access to realistic, interactive 3D tumor cases that can be explored from any angle, providing superior spatial understanding compared to traditional textbook diagrams or 2D images.

Patient Communication: Patients and their families will benefit from intuitive visualizations that clearly demonstrate tumor location, size, and relationship to critical brain structures.

Clinical Workflow: Integrated into the neurosurgical planning pipeline, this system delivers rapid visualization tools that serve as a robust augmentation to traditional imaging modalities

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